

Polar Coordinates & Polar Graphs

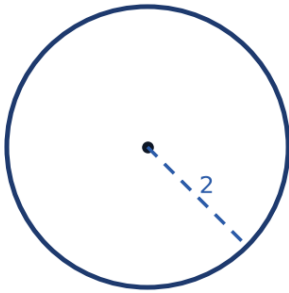


Converting between polar and rectangular coordinates

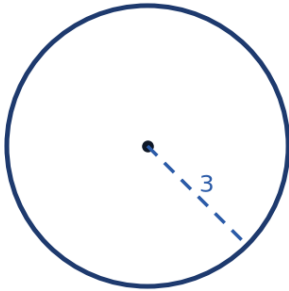
- 1 Convert the polar point $\left(4, \frac{2\pi}{3}\right)$ to rectangular coordinates.
- 2 Convert the rectangular point $(-3, 3)$ to polar coordinates with $r > 0$ and $0 \leq \theta < 2\pi$.

Converting polar equations to rectangular form

- 3 Convert the polar equation $r = 4\cos\theta$ to rectangular form, and identify the curve.

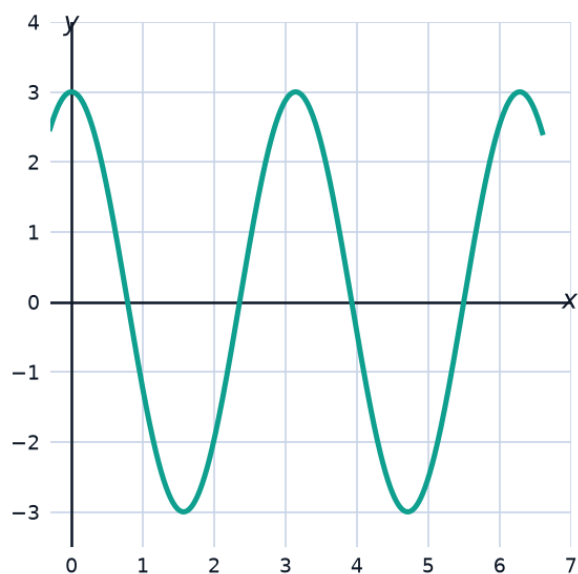


- 4 Convert the rectangular equation $x^2 + y^2 = 9$ to polar form.



Visualizing a rectangular (r vs. θ) graph of a polar rose

- 5 The graph shows $r = 3\cos(2\theta)$ plotted in rectangular form, with θ on the horizontal axis (in radians) and r on the vertical axis — a standard technique for analyzing a polar rose curve before sketching it.

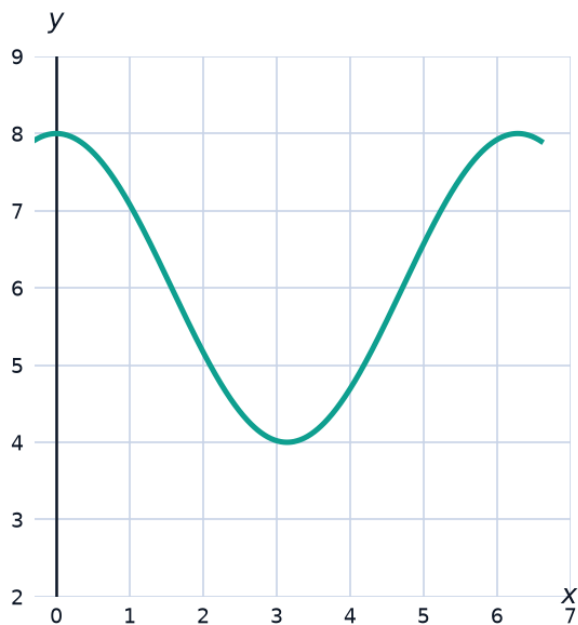


Multiple representations of the same polar point

- 8 Give two other polar representations of the point $\left(5, \frac{\pi}{4}\right)$: one with a negative r -value, and one with a different positive angle (adding 2π).
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Applications of polar curves

- 9 A radar station models the range of its signal by the polar curve $r = 6 + 2\cos\theta$ (in km), where θ is measured from due east. Find the maximum and minimum detection range, and the direction of each.

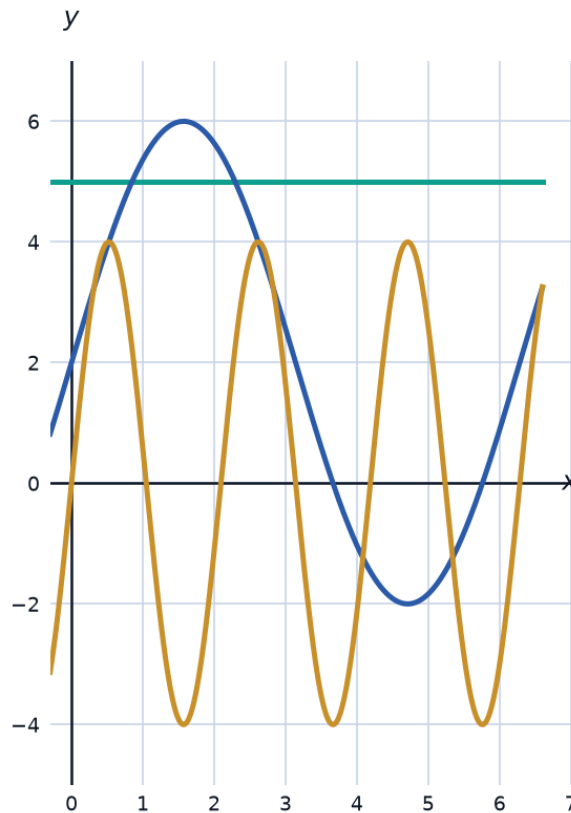


Distance between points in polar form

- 10 Find the distance between the polar points $\left(3, \frac{\pi}{6}\right)$ and $\left(5, \frac{2\pi}{3}\right)$ by first converting both to rectangular coordinates.
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Identifying polar curve families

- 11 Classify each polar equation as a circle, cardioid, limaçon (with inner loop), or rose, and briefly justify: (a) $r = 5$, (b) $r = 2 + 4\sin\theta$, (c) $r = 4\sin(3\theta)$.



Points of intersection of polar curves

- 12 Find the values of θ in $[0, 2\pi)$ where the polar curves $r = 2$ and $r = 4\cos\theta$ intersect.

Area in polar coordinates (introductory)

- 13 Explain, without evaluating an integral, why the polar area formula $A = \frac{1}{2} \int_a^b r^2 d\theta$ uses r^2 rather than r , by comparing it to the area of a circular sector $A = \frac{1}{2} r^2 \theta$.
- 14 A circular sector has radius 6 and central angle $\frac{\pi}{3}$. Use the sector-area formula $A = \frac{1}{2} r^2 \theta$ to find its area.

Multiple choice — easy 15

- 15 Classify the polar curve $r = 3\sin \theta$.
- a. Circle through the pole
 - b. Limaçon with an inner loop
 - c. Cardioid
 - d. Limaçon without an inner loop
- 16 Classify the polar curve $r = 7 + 5\cos \theta$.
- a. Cardioid
 - b. Circle through the pole
 - c. Limaçon with an inner loop
 - d. Limaçon without an inner loop
- 17 Classify the polar curve $r = 2 + 5\cos \theta$.
- a. Limaçon with an inner loop
 - b. Circle through the pole
 - c. Limaçon without an inner loop
 - d. Cardioid
- 18 Classify the polar curve $r = 4 + 6\sin \theta$.
- a. Limaçon with an inner loop
 - b. Cardioid
 - c. Limaçon without an inner loop
 - d. Circle through the pole

19 Classify the polar curve $r = 1 + 4\cos \theta$.

- a.** Limaçon without an inner loop
- b.** Circle through the pole
- c.** Limaçon with an inner loop
- d.** Cardioid

20 Classify the polar curve $r = 8\cos \theta$.

- a.** Limaçon with an inner loop
- b.** Circle through the pole
- c.** Cardioid
- d.** Limaçon without an inner loop

21 Classify the polar curve $r = 1 + 5\sin \theta$.

- a.** Limaçon without an inner loop
- b.** Limaçon with an inner loop
- c.** Cardioid
- d.** Circle through the pole

22 Classify the polar curve $r = 3 + 8\cos \theta$.

- a.** Circle through the pole
- b.** Limaçon with an inner loop
- c.** Cardioid
- d.** Limaçon without an inner loop

23 Classify the polar curve $r = 2 + 2\cos \theta$.

- a. Circle through the pole
- b. Cardioid
- c. Limaçon with an inner loop
- d. Limaçon without an inner loop

24 Classify the polar curve $r = 3 + 3\sin \theta$.

- a. Cardioid
- b. Limaçon without an inner loop
- c. Limaçon with an inner loop
- d. Circle through the pole

25 Classify the polar curve $r = 7\cos \theta$.

- a. Cardioid
- b. Limaçon with an inner loop
- c. Limaçon without an inner loop
- d. Circle through the pole

26 Classify the polar curve $r = 3\sin(5\theta)$.

- a. Rose with 5 petals
- b. Circle through the pole
- c. Cardioid
- d. Rose with 12 petals

27 Classify the polar curve $r = 9\cos \theta$.

- a. Limaçon with an inner loop
- b. Limaçon without an inner loop
- c. Cardioid
- d. Circle through the pole

28 Classify the polar curve $r = 5 + 2\sin \theta$.

- a. Cardioid
- b. Limaçon without an inner loop
- c. Circle through the pole
- d. Limaçon with an inner loop

29 Classify the polar curve $r = 3\sin(3\theta)$.

- a. Rose with 8 petals
- b. Cardioid
- c. Rose with 3 petals
- d. Circle through the pole

Multiple choice — medium 15

30 A circular sector has radius 9 and central angle $\frac{\pi}{3}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

- a. 14π
- b. $\frac{27\pi}{2}$
- c. $\frac{3\pi}{2}$
- d. 27π

31 A circular sector has radius 2 and central angle $\frac{2\pi}{3}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

- a. $\frac{8\pi}{3}$
- b. $\frac{2\pi}{3}$
- c. $\frac{4\pi}{3}$
- d. $\frac{11\pi}{6}$

32 A circular sector has radius 6 and central angle $\frac{\pi}{6}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

- a. $\frac{7\pi}{2}$
- b. $\frac{\pi}{2}$
- c. 3π
- d. 6π

33 A circular sector has radius 4 and central angle $\frac{2\pi}{3}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

- a. $\frac{4\pi}{3}$
- b. $\frac{16\pi}{3}$
- c. $\frac{35\pi}{6}$
- d. $\frac{32\pi}{3}$

34 A circular sector has radius 3 and central angle $\frac{3\pi}{2}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

- a. $\frac{27\pi}{2}$
- b. $\frac{9\pi}{4}$
- c. $\frac{27\pi}{4}$
- d. $\frac{29\pi}{4}$

35 A circular sector has radius 3 and central angle $\frac{3\pi}{4}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

a. $\frac{27\pi}{8}$

b. $\frac{27\pi}{4}$

c. $\frac{31\pi}{8}$

d. $\frac{9\pi}{8}$

36 A circular sector has radius 7 and central angle $\frac{3\pi}{4}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

a. $\frac{151\pi}{8}$

b. $\frac{147\pi}{8}$

c. $\frac{147\pi}{4}$

d. $\frac{21\pi}{8}$

37 A circular sector has radius 4 and central angle $\frac{5\pi}{6}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

a. $\frac{5\pi}{3}$

b. $\frac{20\pi}{3}$

c. $\frac{40\pi}{3}$

d. $\frac{43\pi}{6}$

38 A circular sector has radius 5 and central angle $\frac{\pi}{3}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.

a. $\frac{14\pi}{3}$

b. $\frac{25\pi}{6}$

c. $\frac{25\pi}{3}$

d. $\frac{5\pi}{6}$

- 39 A circular sector has radius 9 and central angle $\frac{5\pi}{6}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.
- a. $\frac{137\pi}{4}$
 - b. $\frac{135\pi}{4}$
 - c. $\frac{135\pi}{2}$
 - d. $\frac{15\pi}{4}$
- 40 A circular sector has radius 5 and central angle $\frac{\pi}{2}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.
- a. $\frac{25\pi}{4}$
 - b. $\frac{25\pi}{2}$
 - c. $\frac{27\pi}{4}$
 - d. $\frac{5\pi}{4}$
- 41 A circular sector has radius 5 and central angle $\frac{\pi}{4}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.
- a. $\frac{5\pi}{8}$
 - b. $\frac{25\pi}{4}$
 - c. $\frac{29\pi}{8}$
 - d. $\frac{25\pi}{8}$
- 42 A circular sector has radius 3 and central angle π . Use $A = \frac{1}{2}r^2\theta$ to find its area.
- a. 5π
 - b. $\frac{9\pi}{2}$
 - c. $\frac{3\pi}{2}$
 - d. 9π

- 43 A circular sector has radius 5 and central angle $\frac{5\pi}{6}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.
- a. $\frac{25\pi}{12}$
 - b. $\frac{125\pi}{6}$
 - c. $\frac{131\pi}{12}$
 - d. $\frac{125\pi}{12}$
- 44 A circular sector has radius 8 and central angle $\frac{\pi}{2}$. Use $A = \frac{1}{2}r^2\theta$ to find its area.
- a. 16π
 - b. 32π
 - c. 2π
 - d. $\frac{33\pi}{2}$
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Multiple choice — hard 15

- 45 Convert the polar point $(-6, \pi)$ to rectangular coordinates.
- a. $(0, 6)$
 - b. $(7, 0)$
 - c. $(6, 0)$
 - d. $(-6, 0)$
- 46 Convert the polar point $(8, \frac{3\pi}{2})$ to rectangular coordinates.
- a. $(-8, 0)$
 - b. $(0, -8)$
 - c. $(0, 8)$
 - d. $(0, 9)$

- 47 Convert the polar point $(2, 0)$ to rectangular coordinates.
- a. $(-2, 0)$
 - b. $(2, 0)$
 - c. $(0, 2)$
 - d. $(3, 0)$
- 48 Convert the polar point $(6, \frac{3\pi}{2})$ to rectangular coordinates.
- a. $(0, -6)$
 - b. $(-6, 0)$
 - c. $(0, 7)$
 - d. $(0, 6)$
- 49 Convert the polar point $(-5, \pi)$ to rectangular coordinates.
- a. $(-5, 0)$
 - b. $(0, 5)$
 - c. $(6, 0)$
 - d. $(5, 0)$
- 50 Find the values of θ in $[0, 2\pi)$ where the polar curves $r = 3$ and $r = 6\cos \theta$ intersect.
- a. $\theta = \frac{2\pi}{3}$ or $\theta = \frac{4\pi}{3}$
 - b. $\theta = \frac{\pi}{4}$ or $\theta = \frac{7\pi}{4}$
 - c. $\theta = \frac{\pi}{6}$ or $\theta = \frac{11\pi}{6}$
 - d. $\theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$

51 Find the values of θ in $[0, 2\pi)$ where the polar curves $r = -2$ and $r = 4\sin \theta$ intersect.

a. $\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$

b. $\theta = \frac{7\pi}{6}$ or $\theta = \frac{11\pi}{6}$

c. $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$

d. $\theta = \frac{\pi}{4}$ or $\theta = \frac{3\pi}{4}$

52 Find the values of θ in $[0, 2\pi)$ where the polar curves $r = 2$ and $r = 4\sin \theta$ intersect.

a. $\theta = \frac{\pi}{4}$ or $\theta = \frac{3\pi}{4}$

b. $\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$

c. $\theta = \frac{7\pi}{6}$ or $\theta = \frac{11\pi}{6}$

d. $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$

53 Find the values of θ in $[0, 2\pi)$ where the polar curves $r = 3$ and $r = 6\sin \theta$ intersect.

a. $\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$

b. $\theta = \frac{7\pi}{6}$ or $\theta = \frac{11\pi}{6}$

c. $\theta = \frac{\pi}{4}$ or $\theta = \frac{3\pi}{4}$

d. $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$

54 Find the values of θ in $[0, 2\pi)$ where the polar curves $r = 1$ and $r = 2\sin \theta$ intersect.

a. $\theta = \frac{7\pi}{6}$ or $\theta = \frac{11\pi}{6}$

b. $\theta = \frac{\pi}{4}$ or $\theta = \frac{3\pi}{4}$

c. $\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$

d. $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$

55 Give an equivalent polar representation of $(3, \frac{\pi}{6})$ using a negative r -value.

a. $(-3, \frac{\pi}{6})$

b. $(3, \frac{13\pi}{6})$

c. $(3, \frac{7\pi}{6})$

d. $(-3, \frac{7\pi}{6})$

56 Give an equivalent polar representation of $(8, \frac{\pi}{2})$ using a negative r -value.

a. $(8, \frac{5\pi}{2})$

b. $(-8, \frac{3\pi}{2})$

c. $(8, \frac{3\pi}{2})$

d. $(-8, \frac{\pi}{2})$

57 Give an equivalent polar representation of $(2, \frac{\pi}{4})$ using a negative r -value.

a. $(-2, \frac{5\pi}{4})$

b. $(2, \frac{5\pi}{4})$

c. $(2, \frac{9\pi}{4})$

d. $(-2, \frac{\pi}{4})$

58 Give an equivalent polar representation of $(9, \frac{3\pi}{4})$ using a negative r -value.

a. $(-9, \frac{3\pi}{4})$

b. $(9, \frac{11\pi}{4})$

c. $(-9, \frac{7\pi}{4})$

d. $(9, \frac{7\pi}{4})$

59 Give an equivalent polar representation of $(7, \frac{5\pi}{6})$ using a negative r -value.

a. $(7, \frac{11\pi}{6})$

b. $(-7, \frac{5\pi}{6})$

c. $(7, \frac{17\pi}{6})$

d. $(-7, \frac{11\pi}{6})$